

AST1410 (Winter 2026) MESA Practical 2

Due 11:59pm 22 Feb 2026

Practicals must be submitted to essick@cita.utoronto.ca before the deadline.

Practicals should be submitted as a PDF describing your results.

Your PDF must be accompanied by a plain ASCII file with all scripts, etc that you developed or a link to a public github repository. If you choose to submit as a github repository, you must specify which commit/hash and branch you are submitting.

Part 1 (5 points)

Read in the data from Practical 1 and make some plots. You may find the following documentation helpful.

- https://docs.mesastar.org/en/latest/using_mesa/output.html

Specifically, plot the star's evolutionary path in the following ways

- an HR diagram (i.e., $\log(L)$ vs $\log(T)$)
- central pressure vs. central density
- central temperature vs. central density

Part 2 (10 points)

Change the convective mixing length parameter and rerun your stellar models.

In ``$MESA_DIR/star/defaults/controls.defaults``, you should find the following description

```
! mixing_length_alpha
```

```
! ~~~~~
```

```
! The mixing length is this parameter times a local pressure scale height.
```

```
! To increase R vs. L, decrease ``mixing_length_alpha``.
```

```
! ::
```

```
mixing_length_alpha = 2
```

Create 2 new stars (in new directories) and update their ``inlist_project`` to declare ``mixing_length_alpha = 1.0`` and ``mixing_length_alpha = 4.0``. Evolve both those stars from pre-main sequence to the same post-main sequence stopping criterion as in Practical 1.

Make the same 3 plots as in Part 1 for the new stars.